

Extrauterine Pregnancy 08. Surgery and Fertility: The Tube That Matters

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 8. Surgery and Fertility: The Tube That Matters.

When surgery is chosen for tubal ectopic pregnancy, the operation has two purposes. The immediate purpose is source control: remove the ectopic pregnancy, stop bleeding, and protect the patient from rupture or ongoing hemorrhage. The second purpose is reproductive: preserve as much future fertility as can be preserved without leaving the patient with avoidable persistent trophoblast, repeat ectopic risk, or an unsafe tube.

The names of the operations matter because they describe different biologic consequences. Salpingectomy removes part or all of the affected fallopian tube. Salpingotomy, also called salpingostomy in many sources, opens the tube, removes the visible ectopic pregnancy, and leaves the tube in place. Salpingotomy preserves anatomy, but it may leave microscopic trophoblast in the tubal wall. Salpingectomy is more definitive for that tube, but it removes that tube from future spontaneous conception.

Older intuition favored tube-preserving surgery for patients who wanted another pregnancy. The intuition is understandable: two tubes sound better than one. The ESEP trial tested that assumption in the exact group where it mattered most: women with tubal pregnancy and a healthy contralateral tube. In that Lancet randomized trial, cumulative ongoing pregnancy by natural conception was 60.7 percent after salpingotomy and 56.2 percent after salpingectomy. The fecundity rate ratio was 1.06, with a confidence interval crossing no effect. Salpingotomy did not significantly improve future natural conception in that population.

Hold the patient population together with the result. ESEP was not every ectopic pregnancy. It was tubal pregnancy with a healthy opposite tube. In that setting, the question was whether preserving the affected tube adds enough fertility to justify its costs. The answer was no significant fertility gain: about 61 percent ongoing natural pregnancy after salpingotomy versus about 56 percent after salpingectomy. The

difference sounds emotionally meaningful until the trial asks whether it is reliable and clinically worth the added burden.

The harms moved in the opposite direction. Persistent trophoblast occurred in 7 percent after salpingotomy and in less than 1 percent after salpingectomy. Repeat ectopic pregnancy was 8 percent after salpingotomy and 5 percent after salpingectomy, a difference that was not statistically definitive but pointed in the direction expected when a previously diseased tube remains. Twenty percent of women assigned to salpingotomy were converted to salpingectomy during the initial operation because of persistent bleeding. Those details matter because they show what "preserving the tube" can cost during and after surgery.

Repeat the harm side in clinical language. Salpingotomy can leave trophoblast behind, so hCG follow-up becomes mandatory. Salpingotomy can leave a damaged tube behind, so recurrence remains a concern. Salpingotomy can fail technically when bleeding persists, so one in five assigned patients in ESEP still required salpingectomy during the operation. Tube preservation is not free simply because the incision is smaller than removal.

The reason this trial changed practice is not that salpingotomy is never useful. It is that a healthy opposite tube changes the fertility calculation. If the other tube is open and functional, one well-functioning tube may be enough for timely natural conception. Keeping a damaged affected tube does not necessarily add useful fertility. It may instead add persistent trophoblast surveillance and a possible site for recurrence.

That is why ACOG and NICE generally prefer salpingectomy when the contralateral tube is healthy. ACOG states that the choice between salpingostomy and salpingectomy should be guided by clinical status, desire for future fertility, and the extent of fallopian tube damage. Salpingectomy is preferred when severe damage is seen or there is significant bleeding from the operative site. It can also be considered in a patient desiring future fertility if the opposite tube is healthy.

The contralateral tube is therefore not a side comment in the operative note. It is central evidence. At laparoscopy the surgeon must inspect the pelvis: the opposite tube, adhesions, hydrosalpinx, endometriosis, prior surgical damage, and the condition of the affected tube. Farren emphasizes that counseling should include age, reproductive history, ultrasound and surgical findings, endometriosis, adhesions, and contralateral tubal damage. Novak states the same principle in clinical language: the decision depends on fertility goals, the clinical picture, and the condition of both the affected and the contralateral tubes.

Salpingotomy becomes more attractive when the patient desires future fertility and the contralateral tube is damaged, absent, or unlikely to support spontaneous conception. In that situation, removing the affected tube may remove the patient's remaining practical route to natural pregnancy. Conservative surgery may be considered if the ectopic is unruptured, the affected tube appears otherwise

salvageable, hemostasis can be achieved without destroying the tube, and the patient understands the follow-up obligations.

This is the same evidence applied in the opposite direction. If the other tube is not healthy, ESEP cannot be used as a simple command to remove the affected tube. The patient may be choosing between a risky preserved tube and no realistic natural tubal pathway. That is why the contralateral tube is not a formality. It changes what "fertility preserving" means.

Slow down the word "tube" during counseling. A fallopian tube is useful only if it can pick up the oocyte, permit sperm and oocyte to meet, move the early embryo toward the uterus, and avoid becoming the implantation bed again. A scarred tube left after salpingotomy may be present on the anatomy diagram but biologically poor. A healthy contralateral tube can preserve spontaneous conception even after salpingectomy. This is why the operation is not simply conservative versus radical. The question is whether the tissue left behind can actually perform the work that fertility requires.

The contraindications to tube-preserving surgery are also mechanical. A ruptured tube may not be reconstructable. A severely damaged tube may preserve tissue without preserving function. Extensive cautery needed for hemostasis can injure the tubal wall enough that the remaining tube is not useful. A recurrent ectopic in the same tube suggests that the tube itself is a persistent risk environment. In these settings, salpingotomy can become a cosmetic fertility operation: it leaves a tube behind but not necessarily a safe or functional one.

Persistent trophoblast is the central postoperative issue after salpingotomy. The visible ectopic pregnancy is removed through an incision, but trophoblast can remain in the tubal muscularis or on peritoneal surfaces. Novak reports persistent ectopic pregnancy after laparoscopic linear salpingostomy in the range of 4 to 15 percent, and the ESEP trial observed persistent trophoblast in 7 percent after salpingotomy. Diagnosis is made when hCG plateaus or fails to resolve. This is why serial hCG follow-up is mandatory after salpingotomy. If persistence occurs and the patient is stable, methotrexate can often treat the residual trophoblast; otherwise surgery may be needed.

The operative route is a separate decision from the procedure. Laparoscopy is preferred when the patient is stable and visualization is adequate, because it is associated with shorter hospitalization, less blood loss, faster recovery, and fewer adhesions than laparotomy. Laparotomy is reserved for instability, large hemoperitoneum, inability to visualize safely, or the need for rapid hemorrhage control. In a bleeding patient, the best fertility-preserving act is control of hemorrhage and survival. Fine fertility optimization belongs only after the patient is safe enough for it.

Counseling should be precise because patients often hear "removing a tube" as "losing fertility." For a patient with a healthy opposite tube, the counseling can be direct: in the best randomized evidence, keeping the affected tube did not

significantly improve future natural pregnancy, but it did increase persistent trophoblast. For a patient with an abnormal or absent opposite tube, the counseling changes: preserving the affected tube may carry real reproductive value, but only if the tube can be preserved as a functioning tube and the patient accepts the surveillance and recurrence risks.

This is where the operation should be explained in whole-system terms. Fertility is not the number of tubes visible at the end of surgery. Fertility is ovulation, pickup, tubal transport, sperm-egg meeting, embryo transport, and safe implantation. A tube that is badly damaged, bleeding, burned by cautery, or already the site of repeated ectopic pregnancy may add anatomy without adding function. A single healthy tube may be more useful than two tubes when one of them is unsafe.

There is also future-pregnancy counseling after any ectopic treatment. ACOG gives a recurrence risk of about 10 percent after one ectopic pregnancy and more than 25 percent after two or more. Nature Reviews Disease Primers gives recurrence estimates that vary by prior treatment and probably reflect underlying tubal pathology as much as the operation itself. The patient should be told that most future pregnancies can still be normally sited, but the next positive pregnancy test should prompt early contact, serial hCG when needed, and ultrasound to confirm location.

So the ESEP result should not be reduced to "always remove the tube." The teachable rule is narrower and stronger: when the opposite tube is healthy, salpingectomy usually avoids persistent trophoblast without sacrificing a proven fertility advantage. When the opposite tube is not healthy, the discussion changes. Good surgical judgment lives in that distinction.

The surgical habit to install is this: do not ask only whether the affected tube can be kept. Ask whether keeping it preserves reproductive function at an acceptable cost. The answer comes from the patient's stability, the bleeding, the affected tube, the opposite tube, fertility goals, and the ability to complete postoperative hCG follow-up.